Fredrik Willumsen Haug

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Education

HARVARD UNIVERSITY Cambridge, MA

S.B. in Engineering Sciences, Bioengineering, Electrical and Computer Engineering (ECE) Track GPA 3.849/4.0 May 2027

Relevant Coursework: Bioengineering (ES53, ES91r, BE110, BE129, ES221, ES227), Electrical Engineering (ES50, ES152, ECE154*, ECE255*), Physics, Mathematics, Chemistry, Biology, Computer Science, Neuroscience. *Spring 2027.

Skills: Programming: C/C++, MATLAB, Rust, Python, R, SQL, TypeScript; Circuit Design: LTspice, KiCad, Breadboard Layout; Tools & Frameworks: Arduino, Raspberry Pi, Git, ImageJ, React, Supabase, Vercel; Laboratory: Cell Culture, Electrophoresis, PCR, Confocal Microscopy, Western Blot.

Publications

POSTER PRESENTATION, HARVARD DATA SCIENCE INITIATIVE August 2026

- '5xFAD Single Unit Encoding Analysis' (first author) — presented at Harvard SPUDS poster session

SCIENTIFIC PUBLICATION, FRONTIERS IN MEDICINE July 2025

- 'Potential source of bias in AI models: lactate measurement in the ICU in sepsis patients as a template' (co-first author) — PubMed PMID: 40703300

ABSTRACT PUBLICATION, AMERICAN THORACIC SOCIETY May 2024

- 'Association of P50 With Physiologic Variables and Outcomes in Critically Ill Patients' — DOI: 10.1164/ajrccm-conference.2024.209.1_MeetingAbstracts.A2959

OPINION ARTICLE, DAGENS MEDISIN December 2023

- 'Når kunstig intelligens diskriminerer' (When Artificial Intelligence Discriminates) — Norwegian healthcare trade press

Research Experience

ENGINEERING SENIOR THESIS, HARVARD SEAS April 2026 - Now

- Designing a battery-powered wireless headstage for neural recordings in rodents (Intan RHD2164 analog front end, nRF54LM20B SoC, Bluetooth LE) that classifies spikes on-device and transmits only unit IDs and timestamps.
- Wrote hardware-timed acquisition firmware in C (timer-triggered SPI with DMA); designed host board PCB; validated the chain end-to-end against a neural simulator with lossless packet transmission and detection of ground-truth spikes.

ALZHEIMER'S DISEASE RESEARCH, HARVARD BIOELECTRONICS GROUP September 2025 - August 2026

- Ran neural recording experiments in 5xFAD Alzheimer mice with subsequent spike-sorting analysis.
- HDSI Research Fellow (summer 2026); analyzed neural representations; built probe drift-correction pipeline.

BIOENGINEERING INTERN (ALZHEIMER'S DISEASE), ROCHE May 2024 - August 2024

- Initiated and maintained microglia cell cultures; tested Roche's AD compounds; conducted imaging and data analysis.

CLINICAL MACHINE LEARNING RESEARCH, MIT IMES September 2022 - September 2024

- Co-designed study on bias in clinical AI using the MIMIC dataset; earned co-first authorship on the publication.
- Developed open-source PyPI package (care-phenotype-analyzer) for healthcare fairness evaluation.

Technical Experience and Teaching

TEACHING ASSISTANT (ECE152), HARVARD SEAS September 2026 - Now

- Hold five hours of problem-solving sessions per week helping students with analog circuit analysis.

SOFTWARE ENGINEERING, FSN CAPITAL September 2024 - August 2025

- Worked as a software engineer in private equity during a college gap year; built a Monte Carlo ROI-distribution model.
- Co-developed full-stack app used by FSN's investment professionals to present deals to the investment committee.

TEACHING ASSISTANT (BST209), HARVARD SCHOOL OF PUBLIC HEALTH August 2023

- Presented a lecture on ensemble models and decision trees; held office hours for statistics and machine learning.

Projects

MULTIMODAL GASTROINTESTINAL SENSOR (ES227 Capstone) Spring 2026

- Electrical engineering lead on team building a multimodal intragastric sensor recording mucosal electrophysiology, contractility strain, intraluminal pressure, and VOC gas composition.
- Designed strain-gauge/AD623 analog front end with Teensy logger; validated in-vivo slow-wave detection.

REAL-TIME BCI SIGNAL PROCESSING LIBRARY (Personal Project) December 2025 - Now

- Developing open-source Rust library for BCI signal processing focused on memory safety and real-time performance.

AI-POWERED BIOTECH EQUITY RESEARCH PLATFORM (CS1060 Capstone) Fall 2025

- Built full-stack platform aggregating ClinicalTrials.gov and PubMed data with AI-driven buy/sell signal detection.

EMG-CONTROLLED ROBOT (ES50 Capstone) Spring 2024

- Built wireless robot driven by EMG biosignals via Arduino Cloud IoT with H-bridge PWM motor control.

LOCATION-BASED SOCIAL MAP (CS50 Capstone) Spring 2023

- Co-built Flask/SQLite social map with live location sharing, activity labels, events, and privacy-first design.